

Myanmar Earthquake & Dam Risk Report

1. Executive Summary

This report covers 9,405 earthquake events in Myanmar from 1970-07-29 to 2026-07-01. The largest event was M7.7 on 1988-11-06. The Gutenberg-Richter b-value is 1.049 with magnitude of completeness $M_c=4.7$.

Total Events: 9,405

Date Range: 1970-07-29 to 2026-07-01

Max Magnitude: M7.7

Mean Depth: 24.8 km

b-value: 1.049

a-value: 7.715

M_c (completeness): 4.7

2. Earthquake Map

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3. Seismic Clustering Analysis

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4. Dam Seismic Risk Assessment

Risk assessment for 254 dams using PGA estimation from the M7.7 mainshock, fault proximity, and structural exposure scoring.

Critical Risk: 25 dams

High Risk: 148 dams

Moderate Risk: 67 dams

Low Risk: 14 dams

Top 30 Highest Risk Dams:

Name	River	PGA(g)	Dist	Fault	Risk	Grade			
Dam near 22.09N 95.86E		nan	0.4019	5.7 km	8.3	Critical			
Dam near 21.77N 95.89E		nan	0.4626	2.7 km	8.3	Critical			
Natkar	nan	0.3593	7.9 km	8.2	Critical				
Dam near 21.47N 95.87E		nan	0.3944	6.0 km	8.2	Critical			
Ta Nai Hka	Ta Nai Hka	0.4441	3.9 km	8.2	Critical				
Dam near 20.72N 95.90E		nan	0.4000	5.6 km	8.2	Critical			
Dam near 21.74N 95.85E		nan	0.3656	7.4 km	8.2	Critical			
Taungnyo	Taungnyo	0.5047	1.2 km	8.2	Critical				
Dam near 20.60N 95.90E		nan	0.3681	7.4 km	8.2	Critical			
Dam near 20.47N 96.08E		nan	0.3252	9.9 km	8.2	Critical			
Dam near 20.85N 96.03E		nan	0.3478	8.3 km	8.2	Critical			
Meikhtila	nan	0.3525	8.2 km	8.2	Critical				
Dam near 17.55N 96.37E		nan	0.3298	9.6 km	8.1	Critical			
Dam near 21.05N 95.83E		nan	0.3105	11.0 km	8.0	Critical			
Dam near 21.04N 95.83E		nan	0.3122	10.9 km	8.0	Critical			
Thaphan Chaung	Thaphan Chaung	0.3041	11.5 km	7.8	Critical				
Dam near 21.34N 95.81E		nan	0.2956	12.2 km	7.7	Critical			
Kun Chaung	Kun Chaung	0.5304	0.9 km	7.6	Critical				
Phyu Chaung	Pyu Chaung	0.5211	1.3 km	7.5	Critical				
Kabaung	Kabaung Chaung	0.3302	9.2 km	7.5	Critical				
Dam near 21.37N 95.80E		nan	0.2830	13.4 km	7.5	Critical			
Yenwe	Ye Nwe Chaung	0.4562	3.9 km	7.4	Critical				
Nyaunggone	nan	0.2795	13.7 km	7.4	Critical				
Dam near 20.73N 95.83E		nan	0.2809	13.6 km	7.4	Critical			
Zeedaw	Nyaung Oat	0.2633	15.3 km	7.2	Critical				
Male-Nattaung	Male-Nattaung	0.2327	18.6 km	6.9	High				
Kyee Ni	nan	0.2362	19.4 km	6.8	High				
Kintat	Mu	0.2235	21.0 km	6.7	High				
Dam near 20.58N 96.17E		nan	0.2281	20.7 km	6.7	High			
Myohta	Ku Htaung	0.2211	20.6 km	6.7	High				

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5. Aftershock Forecast

Modified Omori law parameters fitted to aftershocks of M7.7 mainshock.

K (productivity): 4.53

c (time offset): 0.100 hours

p (decay exponent): 0.932

Expected M $\geq$ 3 (7d): 0

Expected M $\geq$ 3 (30d): 0

Expected M $\geq$ 3 (90d): 0

Note: These are statistical estimates based on the Omori decay law. Actual aftershock rates may vary due to local geological conditions.

6. Data Sources

- Earthquake data: mmeq.akze.net API (USGS/ISC sourced)
- Dam data: Open Development Mekong (IFC/WLE), CC BY-SA 4.0
- Fault lines: USGS/Plate boundary data
- PGA estimation: Abrahamson & Silva (2008) NGA-West1 GMPE
- Report generated by mmeq-opendata toolkit